

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING AND TECHNOLOGY, KATTANKULATHUR CAMPUS
DEPARTEMNT FO MECHANICAL ENGINEERING



Course Code	21EIC553J	Course Website:	
Course Title	ROBOT PROGRAMMING	Instructor:	Dr. A. Xavier Reni Prasad
Semester	EVEN 2025-2026	Email:	xavierra@srmist.edu.in
Pre-requisite / Co-Requisite	NIL	Slot & Class Hours:	Slot B Day Order 2 & Day Order 3
Lecture Hall #	ESB409 - ELEC.SCIENCE	Laboratory Location	Robotics & Factory Automation Lab

Organisation of the course

Classes will be predominatntly lectures with technology advancements and discussion on R&D works. Guest Lecturers from R&D are invited

Course Objectives and Learning Outcome(s)

- 1 Introduces students to foundational concepts in robot programming
- 2 Covers advanced topics in robot autonomy, emphasizing slam for mapping and localization, techniques
- 3 Explores topics in multi-robot systems, object tracking, and human-robot interaction
- 4 Addresses advanced topics in human-robot interaction HRI), dynamic path planning, and advanced kinematics.

Learning Materials

1. Morgan Quigley, Brian Gerkey, and William D. Smart" Programming Robots with ROS", 1st Edition, O'Reily Media, 2015
2. Roland Siegwart, Illah R. Nourbakhsh, and Davide Scaramuzza, "Introduction to Autonomous Mobile Robots", 2nd Edition, MIT Press, 2011
3. Adrian Kaehler and Gary Bradski, "Learning OpenCV 3: Computer Vision in C++ with the OpenCV Library", 1st Edition, O'Reily Media, 2016
4. Mark W. Spong, Seth Hutchinson, and M. Vidyasagar, "Robot Modeling and Control", 1st Edition, Wiley, 2005
5. Cameron Hughes, Tracey Hughes, "Robot Programming: A Guide to Controlling Autonomous Robots", Pearson Education, 2016

Course Topics and Schedule

Week	Major Topic	Module / Unit	Method of delivery (Multi-Select)	Take home Assignment(s)	Learning materials to be referred
1	Introduction to ROS	1	Lecture/Presentation		L1
2	Basic Robot Control	1	Lecture/Video		L4
3	Line Following Algorithm	1	Lecture/Presentation		L1
4	RoboAnalyzer	1	Individual tasks		L4

5	Basic movement, Obstacle avoidance & Line following robot	1	Group activity	Assignment # 1	
6	SLAM (Simultaneous Localization and Mapping)	2	Lecture/Presentation		L2
7	Localization Techniques	2	ped mode Discussion / Debate		L2
8	Vision Based Navigations	2	ped mode Discussion / Debate		L2
9	Mapping and localization, Autonomous navigation & Vision based navigation	2	aticum / in-class assignm	Assignment # 2	L1,L2,L3
10	Multi-Robot Systems	3	Lecture/Presentation		L5
11	Robotic Arm Control	3	Lecture/Video		L2
12	Object Tracking and Following	3	Lecture/Presentation		L3
13	Multiple robot coordination, Pick and Place with robotic arm & Follow the Leader	3	Practicum / in-class assignment	Assignment # 3	L5,L3
14	Human Robot Interaction	4	Flipped mode Discussion / Debate		L5
15	Dynamic Path Planning	4	Individual tasks		L5
16	Advanced Kinematics	4	Lecture/Presentation		L4
17	Human Robot Interaction using voice and gesture	4	Practicum / in-class assignment	Assignment # 4	L4,L5
18	Path planning in dynamic environment	4	Practicum / in-class assignment	Assignment # 5	L4,L5
19	Joint inverse kinematics for complex robot config	4	Practicum / in-class assignment	Assignment # 6	L4,L5

Grading Criteria

Type of assessments	Periodicity	Weightage	Due Date (if applicable)	Remarks
FJ1-Test-10 Marks FJ2-Test-20 Marks	After 4th and 11th week respectively	30%	5-3-2026 to 12-3-2026; 29-4-2026 to 8-5-2026	Written Exam
FJ1-Assignment 1&2 -5 Marks + Review 1 FJ2-Assignment 3 to 6 -10 Marks + Review 2	Monthly once	15%	5-3-2026 to 12-3-2026; 23-3-2026 to 17-4-2026; 29-4-2026 to 8-5-2026	Sum of all

Practical & Viva-Voce) - 5 Marks, Project Demo / Video Submission/Report/ Paper Presentation - 10 Marks	After 11th week	15%	29-4-2026 to 8-5-2026	Sum of all	
Continuous Assessment	Total	60%			
End Semester Examination		40%			
	Total	100%			

Make-up Policy

Assignments / Term Project

Proper academic performance depends on students doing their work not only well, but on time. Accordingly, assignments for this course must be received on the due date specified for the assignment. No Make-up will be available for assignments or term project. Late submissions will be evaluated at 25% less weight for that component for a delay of up to 24 hours after which no submissions will be accepted.

Test

Prior Permission of the course coordinator is usually required to get a make-up test.

A make-up test shall be granted only in genuine cases where, in the course faculty's judgment, the student would be physically unable to appear for the test.

Course faculty's decision in this matter would be final.

Academic Integrity

Plagiarism or cheating will result in a grade of zero for the assignment or exam. Severe cases may result in course failure.

Additional Information

Provide a brief description (e.g. field trips, special lab session, special tutorials, industry certifications etc), dates, times, required materials or preparation, any fees or costs, etc.

Additional academic support

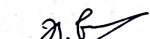
If any student requires additional free academic support (additional coaching), he/she may reach out to the Faculty Advisor

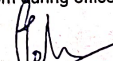
Notices

All notices concerning this course will be displayed online only. If there is a need srmist email would be used.

Contact Information

Feel free to reach out to Dr A. Xavier Reni Prasad via email for any questions regarding the course material. For urgent matters, you can visit the faculty room during office hours.


Course Faculty


HOD